

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

9th September 2021

Subject CM1A – Actuarial Mathematics
(Paper A)

Time allowed: 3 Hours 30 Minutes (9.30 - 13.00 Hours)

Total Marks: 100

- Q.1)** A fixed interest bond with a term of 10 years pays 7% coupon semi-annually in arrears. The gross redemption yield for the bond is 9% p.a. The nominal value of the bond is INR 100. Select the option which best represents the market value of this bond.
- a) 88.15
b) 65.19
c) 89.14
d) 134.1
- [4]**
- Q.2)** Describe the risks covered in a typical car insurance policy. Describe the cashflows from the perspective of the policyholder.
- [5]**
- Q.3)** Define the following:
- i) $a_{\overline{n}|}$ (1.5)
- ii) a_x (1.5)
- iii) Principle of equivalence or equation of value. Mention in terms of loss function as well. (2)
- [5]**
- Q.4)** ABC Company is a new start-up for which employee attrition is an area of concern. The probability of each employee leaving the company during any given year is 7% independent of another employee. To improve its attrition rate, the company has decided to offer a benefit of INR 1,000,000 to employees who remain with the company till the end of 25 years. Select the option which best represents the standard deviation of this benefit payment outgo, assuming an effective rate of interest of 5% and ignoring the possibility of death.
- a) 109,063
b) 82,490
c) 0
d) 109
- [4]**
- Q.5)** A life insurance policy shall pay a benefit of INR 200,000 at the end of year of death to a policyholder aged exactly 50 years, provided that death occurs after the policyholder has attained the exact age of 60 years. Select the option which most closely represents the expected present value of benefits assuming that the effective annual rate of interest is 6% and the mortality follows the AM92 Ultimate table.
- a) 30,000
b) 49,000
c) 59,000
d) 35,000
- [4]**
- Q.6)** A life currently aged x years is subject to constant force of mortality 0.03 per annum. The constant force of interest is 0.05 per annum. Calculate:
- i) a_x (4)
- ii) $\ddot{a}_x$ (1)
- [5]**

- Q.7)** ABC Insurance Company is a leading insurance company selling protection business. They sell decreasing term assurance plans where the benefit terms can be customised to the needs of customers.

An individual aged 55 years wants to guarantee an amount of INR 5,000,000 to his family upon his death. He currently invests INR 20,000 per month into an account that earns an interest at 1% per month. ABC Insurance Company sells a decreasing term assurance plan to him with gross premium of INR 5,000 payable monthly and death benefit being calculated as difference between INR 5,000,000 and the balance in the account at the end of the month.

In respect of the insurance plan, ABC Insurance Company has an initial expense of INR 2000 and a renewal expense of INR 50 per month, incurred from beginning of second policy month inflating at 3% p.a. Further, ABC Insurance Company pays a renewal commission of 5% of premium.

Calculate the expected net cash flow for ABC Insurance Company in the 30th month from this contract. You may assume mortality to be 100% of AM92 Ultimate with an age set back of 3 years and assume uniform distribution of deaths over each year of age. You should ignore any interest income.

[7]

- Q.8)** A life insurance company sells an insurance product with a policy term of 5 years that pays benefits to the customer as described below:

- INR 800,000 at the end of the year of death during policy term
- INR 500,000 at the end of the year of withdrawal due to permanent sickness during policy term
- INR 300,000 upon survival to the end of the policy term
- 60% of the total premiums paid till date at the end of the year upon lapsation

Below assumptions apply:

- Force of mortality = 0.01
- Force of sickness = 0.04
- Force of lapsation = 0.05
- Discount rate of 5% p.a.
- Annual premium of INR 100,000 paid in advance

- i) Calculate the dependent probabilities of death, sickness, lapse and survival. (4)

- ii) Calculate the net present value of the contract ignoring expenses and other charges. (8)

[12]

- Q.9)** The liabilities for an insurance company have been calculated to be worth INR 1500 crores at a rate of interest of 6% p.a. A sensitivity of 10% multiplicative increase in interest rate results in the reserves falling down to INR 1360 crores. Which of the options below best represents the duration of the liabilities of the insurance company:

- a) 9 years
- b) 16 years
- c) 1 year

d) 2 years

[3]

Q.10) A 91-day treasury bill was purchased for INR 103 at the time of issue and later sold to another investor for INR 104 who held the bill to maturity. The bill was redeemable at INR 105. The rate of return received by the initial purchaser was 5% p.a. effective.

i) Choose which option shows the length of time for which the initial purchaser held the bill.

- a) 45 days
- b) 72 days
- c) 55 days
- d) 60 days

(3)

ii) Choose which option shows the annual rate of return achieved by the second investor.

- a) 50%
- b) 19%
- c) 45%
- d) 40%

(3)**[6]**

Q.11) Two lives male and female, both aged 60 years exact purchase a joint whole life policy. The below benefits are payable:

- Lump sum of INR 500,000 payable immediately upon the first death
- An income of INR 400,000 payable annually in advance until the female is alive, starting from next year after the death of the male life
- An income of INR 200,000 payable annually in advance until the male is alive, starting from next year after the death of the female life

Level premiums are payable monthly in advance while both lives are alive, for a maximum of period of 20 years. Calculate the gross premium for this product assuming the below:

Mortality: PMA92C20 and PFA92C20 for the male and female life respectively.

Interest rate: 4% p.a.

Initial Commission: 20% of annual premium incurred at the time of sale of policy

Renewal expenses and commission (together): 3% of monthly premium incurred from inception of policy.

Initial expense: 10% of annual premium incurred at the time of sale of policy.

[10]

Q.12) Pallavi is a student studying in the USA. She funded her two years masters' degree through personal savings and a student loan. The total fee for her education was INR 15,000,000. On 1st March 2015, the student loan was made available to her by ABC Bank at an interest rate of 6.5% p.a. compounded annually. The loan was to be repaid through monthly instalments of INR 200,000 paid in arrears over a period of 5 years, with the first payment being due on 31 March 2017.

- i) Calculate the amount of personal savings put in by Pallavi in March 2015 to fund her education. (4)

On 1st March 2020, Pallavi decided to take out a home loan for INR 10,000,000. ABC bank offered a facility to combine the home loan with her outstanding student loan and also offered better terms. The total loan was to be repaid over a period of 10 years, through annual instalments paid in arrears at an interest rate of 5% p.a. compounded annually.

- ii) Calculate the annual instalments that Pallavi would pay in future. (5)

The other option for Pallavi is to keep the two loans separate. In this case the student loan will continue of the earlier agreed terms. On the other hand, the home loan will need to be repaid over a period of 12 years, through annual instalments paid in advance at an interest rate of 5% p.a. compounded annually.

Pallavi has decided that she will choose the option, which results in a lower cumulative interest amount payable.

- iii) Calculate the cumulative interest amount under both available options and identify the option that Pallavi should choose. (5)
[14]

- Q.13)** An insurer issues a decreasing term assurance combined with an annuity contract to a life aged 30 years exact. Level premiums are payable monthly in advance for a maximum term of 30 years.

On death of the life insured before age 60 years a benefit is paid immediately. The benefit is INR 500,000 on death in the first year of contract, INR 485,000 on death in the second year, INR 470,000 on death in the third year and so on with the benefit decreasing by INR 15,000 each year until age 60 years is attained. No death benefit is payable after age 60 years.

On attaining age 60 years the life receives a whole life annuity of INR 20,000 p.a. payable monthly in arrears.

Calculate monthly premium using the following basis:

Mortality:	upto age 60	AM92 Select
	Over age 60	PMA92C20
Interest	4% p.a.	
Expenses	Initial:	Rs.850 per policy
	Regular:	Rs.50 in the first year, inflating at the rate of 4% p.a., assumed to be incurred annually at the start of each year during the deferment period.
Claim expense:	0.75% of each annuity payment	

[13]

- Q.14)** i) An insurance company is analysing its current policyholders to further strengthen its market reach. In order to do so, it uses all of its policyholder data and calculates the mean and standard deviation for age and annual premium. Select the option which describes the nature of data analysis undertaken:

(2)

- a) Predictive analysis
 - b) Descriptive analysis
 - c) Inferential analysis
- ii)** The company wants to understand the impact on its balance sheet due to decline in interest rates in the market. In order to do so, it assesses the impact of change in yields on assets and correspondingly determines a revised valuation interest assumption for calculation of liabilities. At the same time, it also determines a new inflation assumption. Select the statement which is true:
- a) This is a stochastic model
 - b) This is a scenario-based model
 - c) This is a proxy model
- (2)
- iii)** One of the analysis to be undertaken by the insurance company requires development of Monte Carlo simulations. However, the actuarial team has advised that performing a Monte Carlo simulation is a time-consuming process and so a simpler alternate modelling should be adopted. Select the option which applies:
- a) Simplified model is an example of a proxy model
 - b) Given that Monte Carlo simulations are proposed, this model has a continuous state process
 - c) Given that Monte Carlo simulations are common in actuarial work, limitations on the use and validity of such models are not required to be highlighted to users of this model.
- (2)
- iv)** A company is collecting data on the periods of time for which a user's internet connection was disrupted, but only recorded the duration of periods of disruption that lasted 5 minutes or longer. Select the term which best describes this data set:
- a) Stratified data
 - b) Censored data
 - c) Truncated data
 - d) Big data
- (2)

[8]
